

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid Term Examination, Autumn 2025

Course Title: Machine Learning

Course Code: CSE 431

Time: 1(One) hours

Marks: 20

(Answer all questions)

Marks

1. Table 01 presents an agricultural dataset that records the relationship between rainfall, fertilizer usage, and corresponding crop yield. The objective is to develop a predictive model that estimates crop yield based on the given independent variables (rainfall and fertilizer usage).

Table 01

Rainfall (mm)	Fertilizer Used (kg/acre)	Crop Yield (tons/acre)
200	50	2.5
250	60	3
300	70	3.6
350	80	4.2
400	90	4.8

- a) Define classification and regression in the context of machine learning. [02]
- b) Use stochastic gradient descent to construct a model to fit the data from table 01. The initial parameters are $(\theta_0, \theta_1, \theta_2) = (0, -1, 1)$ and learning rate $\alpha = 0.01$. Apply 2 iterations. [05]
- c) Two regression models are trained to predict crop yield based on rainfall, fertilizer usage, and several newly added features from Table 01, including soil pH level, average sunlight hours per day, irrigation frequency, pest control measures. [03]
- i. Model 1: A simple linear regression model using only rainfall as the predictor.
- ii. Model 2: A highly complex regression model utilizing all available features, including the newly added ones, with a large number of parameters.

Analyze which model is more likely to suffer from underfitting and which is more likely to suffer from overfitting.

2.

A health research team collected a small dataset to predict whether a person is at risk of heart disease based on their cholesterol level and age, which is given as table 03.

Table 02

Age (years)	Cholesterol (mg/dL)	Risk (Y)
40	200	0
50	220	0
60	250	1

- Build** a logistic regression model to classify individuals as high risk (1) and low risk (0). Set initial parameters randomly and show only one iteration. [04]
- Explain** the difference between train, validation and test dataset. [03]
- Assume that the dataset in Table 01 has three labels instead of two: Risk (0), Moderate Risk (1), and No Risk (2). **Describe** how logistic regression can be applied to classify data with multiple labels. [03]

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Program: B.Sc. in CSE

Mid Term Examination, Autumn 2025

Course Title: Internet of Things

Course Code: CSE 401

Marks: 20

Time: 1(one) hour

(Answer all questions)

Q.No.	Questions	Marks
1.	a) Outline Internet of Things (IoT). Discuss the IoT characteristics, explaining advantages and disadvantages of IoT devices.	[05]
	b) Discuss in detail about the technologies that have made Internet of Things (IoT) possible.	[05]
2.	a) Logical design is very significant to understand IoT system to IoT developer. Discuss the terms that are described for understanding the Logical Design of IoT.	[05]
	b) A company wants to build a simple IoT system to monitor the temperature in a single room. The system will use a temperature sensor to collect data, a microcontroller to process it, and a display to show the current temperature to anyone in the room.	[05]
Briefly describe the Internet of Things (IoT) communication models with real life examples. Then, for communication between the sensor and the microcontroller, a specific communication model is most suitable. Identify this model and Explain your reasoning.		

University of Information Technology & Sciences (UITS)
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Department of Computer Science & Engineering
Program: B.Sc. in CSE
Mid Term Examination, Autumn 2025
Course Title: Operating Systems
Course Code: CSE 411

Marks: 20

Time: 1(one) hour

(Answer all questions)

1. a) Suppose a web server is running on a Linux system. It handles thousands of HTTP requests per second. To efficiently manage this load, the server uses a multithreaded model. To manage these threads, the OS uses kernel threads that can run independently on different cores, improving CPU utilization. Each client request is handled by a kernel thread. Since they share the same address space, communication between threads is fast. If one thread blocks, the kernel can schedule another thread on the CPU. [05]

- i. Explain the kernel thread in the context of the Linux operating system.
- ii. Explain how the web server prefers kernel threads over user-level threads.
- iii. Describe the benefits of using kernel threads in this server scenario.

- b) A mid-sized tech company runs a custom-built operating system on its in-house servers to handle multiple user applications, background services, and data processing tasks. The system handles hundreds of processes at a time, including user login sessions, data analytics jobs and file I/O operations. [05]

- i. Explain the role of each of the three key scheduling queues in process management.
- ii. Describe how optimizing scheduling queues can improve overall system performance.

2. a) Some processes have arrived to get CPU service and their details are given in the following table. [07]

Process	Arrival Time	Burst Time
P1	0	16
P2	1	5
P3	2	6
P4	3	1
P5	4	2

Now apply **Preemptive Shortest Job First** scheduling algorithm and find the following-

- i. Gantt chart of execution of the processes.
 - ii. Turnaround Time and Waiting Time of each process.
 - iii. Average Waiting Time for the given scenario with the given algorithm.
- b) Scheduling algorithms are used to give CPU services to the processes. Preemptive [03]
and Non-Preemptive are two different approaches of scheduling the processes.

Explain the differences between preemptive and non-preemptive CPU scheduling algorithms. Give one advantage and one disadvantage of each.

University of Information Technology & Sciences (UITS)

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Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2025

Course Title: Machine Learning

Course Code: CSE 431

Marks: 50

Time: 3 (Three) hours

(Answer all questions)

Marks

1.

A hospital is analyzing patient data to predict the risk of heart disease. The dataset contains information about patients' age group, gender, smoking status, and exercise level, along with whether they are at high or low risk of heart disease (Table 01). Two data scientists are working on this problem:

- Data Scientist A is building a Decision Tree model to classify patients based on the categorical features.
- Data Scientist B is building a Naive Bayes model using the same dataset to predict heart disease risk.

Table 01

Age Group	Gender	Smoking Status	Exercise Level	Heart Disease Risk
40-50	Male	Smoker	Low	High
30-40	Female	Non-Smoker	Medium	Low
50-60	Male	Smoker	Low	High
40-50	Female	Non-Smoker	High	Low
30-40	Male	Smoker	Medium	High
50-60	Female	Non-Smoker	Low	High
40-50	Male	Non-Smoker	Medium	Low
30-40	Female	Smoker	Low	High

a) State the Bayes Theorem formula.

[02]

b) Show how data scientist A calculates the entropy of the whole dataset.

[03]

c) Following the approach of data scientist B, predict whether a patient with the following profile will have a high risk of heart disease:

[05]

Age Group = 40-50

Gender = Female

Smoking Status = Smoker

Exercise Level = Low

2

A city planner is analyzing the locations of eight community facilities, given by the coordinates A1(1,9), A2(3,6), A3(7,3), A4(5,7), A5(8,5), A6(6,2), A7(2,1), and A8(4,8), to identify natural groupings. The goal is to use a clustering algorithm to group the facilities based on spatial proximity, enabling more efficient allocation of resources and optimized service coverage across the area.

- Apply** a suitable clustering algorithm to the given data points to group them into 3 clusters, performing at least 2 iterations to allow the algorithm to converge and identify the natural groupings. [05]
- The locations of the eight community facilities can be transformed into a categorical dataset by labeling each coordinate as High (H) if it is greater than 5 and Low (L) if it is 5 or below. **Apply** K-Modes clustering to group the facilities into 2 clusters, performing at least 2 iterations to identify the natural groupings based on these categorical attributes. [05]

3.

Table 02 describes rainfall and distance-to-river measurements for eight land areas, along with their flood-risk classification as either "At Risk" or "Safe."

Table 02

Rainfall (mm)	Distance to River (km)	Class (At Risk/Safe)
90	1.2	At Risk
110	1.5	At Risk
290	3.9	At Risk
310	3.5	At Risk
130	3.6	Safe
250	1.3	Safe
180	2.4	Safe
200	2.8	Safe

a)

Differentiate between hard and soft margin in Support Vector Machine. [02]

b)

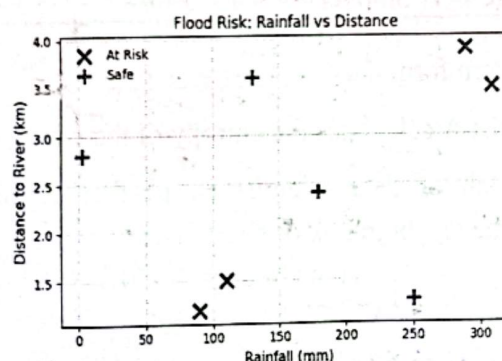


Figure: 01

Figure 01 presents a scatter plot of the given dataset. The plot illustrates that the classes are not linearly separable. **Explain** the concept of feature expansion and its application in enabling Support Vector Machines to address such non-linearly separable data. [03]

- c) Like SVM, KNN aims to separate the classes, but instead of finding an optimal hyperplane, it classifies a point based on the majority class of its nearest neighbors. [05]
 Apply the K-Nearest Neighbors (KNN) algorithm to classify the dataset based on rainfall and distance to the river. Assume $k=3$. (150, 2 km)

4. in a manufacturing quality control system, an ensemble model is deployed to classify components into one of three categories based on sensor and inspection data: Class A, Class B, and Class C. The ensemble combines the outputs of five classifiers—Decision Tree, Logistic Regression, k-NN, SVM, and Random Forest.

For a newly inspected component, the predicted probabilities from each classifier are summarized in Table 03:

Table 03

Classifier	A	B	C
Decision Tree	0.6	0.3	0.1
Logistic Regression	0.45	0.35	0.2
k-NN	0.5	0.2	0.3
Support Vector Machine (SVM)	0.35	0.5	0.15
Random Forest	0.7	0.2	0.1

- a) Define boosting classifiers. [02]
 b) Briefly explain how a Random Forest works and how it uses bagging with randomized data subsets. [03]
 c) Calculate the ensemble's predicted class using soft voting by averaging the predicted probabilities for each class across all five classifiers, then determine the predicted class using hard voting, where each classifier votes for its highest-probability class. [05]

5. A machine learning model is being trained to classify a single channel image into 4 different classes. Here is the pixel values of a sample image of size 5×5 from the dataset.

100	50	0	0	0
0	75	50	0	0
0	25	250	200	0
0	100	175	150	0
100	100	50	10	0

- a) Explain which activation function should be used in the output layer for the given scenario. [03]
 b) Describe the structure of a perceptron. [02]
 c) Apply the following convolution kernel to the input image using a stride of 1. Then, apply the ReLU activation function to the convolution output. Finally, perform 2×2 max pooling with a stride of 1 and show the resulting output. [05]

1	1	1
0	0	0
-1	-1	-1

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Time: 3(three) hours

Marks: 50

(Answer all questions)

Marks

Q. No.

(1. a)

Here are Five processes named as P1, P2, P3, P4, and P5. Their arrival time and burst time are given below in the table. The time quantum of the system is 4 units.

[07]

Process	Arrival Time	Burst Time
P1	0	5
P2	1	6
P3	2	3
P4	3	1
P5	4	5

Table-01: Processes with arrival time and burst time

Apply Round Robin scheduling algorithm and calculate the following:

- Gantt chart of execution of the processes.
- Turnaround Time and Waiting Time of each process.
- Average Waiting Time for the given scenario with the given algorithm.

[03]

(b)

Define the following terms :

- Page fault
- Logical Address
- Physical Address

[07]

(2. a)

Consider six memory partitions of sizes:
100 KB, 500 KB, 200 KB, 400 KB, 600 KB, 300 KB
These partitions need to be allocated to four processes of sizes:
212 KB, 415 KB, 95 KB, 500 KB in that order.

Perform the allocation of processes using-

- First Fit Algorithm
- Best Fit Algorithm
- Worst Fit Algorithm

(b)

Describe the concept of Internal Fragmentation and External Fragmentation with suitable examples of each.

[03]

3. a) One of the most vital mechanisms in OS is the interrupt system. The concept of interrupts is central to the responsiveness, efficiency, and real-time functioning of computer systems.

Answer the following questions:

- Define Interrupt and write the necessity of interrupt in computer system.
- Explain Hardware and Software interrupt with example of each type.

b) Describe how interrupt is handled in operating system with a proper flow chart.

a) Explain the necessary conditions to occur a deadlock.

b) Depict how physical address and logical address is coordinated in Operating System (OS) with suitable diagram.

5. a) Explain the concept of Deadlock with a real world example.

b) Considering a system with five processes P0 through P4 and three resources A, B, C. Resource type A has 10 instances, B has 6 instances and type C has 8 instances. Suppose at time t0 following snapshot of the system has been taken:

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	1	1	2	5	3	4	2	2	2
P1	2	0	1	4	2	3		1	
P2	3	2	2	6	3	3			
P3	2	1	0	3	2	1			
P4	0	1	1	4	2	3			

Table-02: Processes with allocation

Now apply Banker's algorithm and answer the following questions:

- Compute the Need matrix.
- Apply the Safety Algorithm and check if the system is safe.
- Find the safe sequence, if one exists.
- Suppose P4 requests (1,0,1) — can the request be safely granted?

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Course Title: Internet of Things
Course Code: CSE 401

Marks: 50

Time: 3(three) hours

(Answer all questions)

Q.No.	Questions	Marks
1.	A company named "RobotBanai" is developing a fully automated manufacturing system. Their engineers plan to integrate Machine-to-Machine (M2M) communication for robots, sensors, and actuators to improve efficiency and reduce downtime.	
a)	Explain the operation of M2M communication within the factory environment and identify five communication protocols suitable for this industrial setting.	[05]
b)	Compare the M2M architecture used in RobotBanai with an equivalent IoT architecture, highlighting the difference in data flow and connectivity design.	[05]
2.	A healthcare provider, "Medoctor" wants to enhance its hospital network performance to support real-time patient data transmission and video monitoring using SDN and NFV technologies.	
a)	Describe the architecture of SDN for Medoctor's hospital network and explain the role of SDN controller in dynamically managing medical data traffic.	[05]
b)	Analyze the impact of combining SDN and NFV on improving the network's flexibility, security, and cost-efficiency for telemedicine and remote diagnostics.	[05]
3.	An ISP named "Botnet" still relies on SNMP for managing its routers and switches but plans to transition to NETCONF and YANG for cloud-scale network automation.	
a)	Compare SNMP and NETCONF in terms of their management approach, data modeling, and suitability for modern cloud networks.	[05]
b)	Define YANG and evaluate its role enhancing network configuration and automation compared to traditional CLI-based management.	[05]

4.

A startup, "GeoSense" is developing a smart city platform that integrates geospatial data, transportation analytics, and environmental sensors.

- a) **Identify and describe** four common reference architectures applicable to GeoSense's IoT platform. Discuss their significance in system standardization and interoperability. [05]
- b) **Analyze** the functional architecture of the Open Geospatial Consortium (OGC) model for GeoSense's project. [05]

5.

A logistics company, "Naraz" is implementing a cloud-based IoT solution to monitor its fleet in real-time. The system collects data such as GPS, fuel usage, and cargo conditions through IoT sensors.

- a) **Describe** the types of APIs that can be used in Naraz's cloud-IoT integration and explain their respective roles in enabling data communication [05]
- b) Using the context of Naraz's system, **illustrate** the supports provided by the WAMP protocol for real-time peer-to-peer communication between tracking devices, cloud servers, and dashboards. [05]